Activity 3.7 – Instruction Coverage Monitoring

1.Design Approach:

Designed an instruction coverage monitoring module using Verilog HDL to identify different RISC-V instruction categories, including arithmetic, load, store, branch and jump instructions. Implemented opcode-based detection and instruction counting.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Provided different instruction opcode values.
- Tested arithmetic instruction detection.
- Verified branch and load/store opcode detection.
- Checked instruction counter operation.
- Simulated using ModelSim.

3.Results:

The testbench checks opcode identification and instruction counting. PASS/FAIL messages are displayed.

4.Issues Faced:

Opcode values must match the correct RISC-V instruction encoding. Coverage flags must be updated correctly. Counter reset and increment conditions require verification.

5.Conclusion:

Successfully designed and verified an instruction coverage monitoring module using Verilog HDL.